13. [14 marks] *(2014:39)*

An organic compound that contains only carbon, hydrogen, oxygen and bromine was analysed to determine its empirical formula. A combustion analysis of 1.50 g of the compound produced 1.58 g of carbon dioxide and 0.563 g water.

On treatment of 1.75 g of the compound to convert the bromine in the compound to bromide ions and further reaction with silver nitrate, 1.97 g of silver bromide was precipitated.

(a) Determine the empirical formula of the compound. [10]

(b) 1.95 g of the compound was vaporised and was found to occupy 0.387 L at 95.0 kPa and 105°C. Determine the molecular formula of the compound. [2]

(c) Further analysis of the organic compound revealed that it had a carboxylic acid functional group. Draw a possible structural formula of the organic compound. [2]

c) i) How the lead-acid battery produces current:
The electrodes are made out of lead alloy grids packed with spongy lead or finely divided PbO_2. These electrodes are arranged alternatively and suspended in dilute sulfuric acid.
When the circuit is closed during the discharging process, lead at the anode is oxidized to form $PbSO_4$. The anode reaction is: $Pb(s) + SO_4^{2-}(aq) \rightleftharpoons PbSO_4(s) + 2\,e^-$
The electrons released reduce PbO_2 in the presence of H^+ ions and SO_4^{2-} ions. $PbSO_4$ forms on the cathode. The cathode reaction is: $PbO_2(s) + SO_4^{2-}(aq) + 4\,H^+(aq) + 2\,e^- \rightarrow PbSO_4(s) + 2\,H_2O(\ell)$. The result of the discharge is the formation of $PbSO_4$ and a decrease in the concentration of H_2SO_4 which results in the lowering of its density of the electrolyte.
The battery is charged when the car is running and the reactions are reversed during this process.

ii) The electrical potential of the cell is determined by the difference between the two half-cell potentials, the temperature and, the concentration of the electrolytes.

d) i) Given the concentration and volume of the H_2SO_4 solution, the number of moles of H^+ ions in the lead-acid battery is = c × v = 3.55 mol L^{-1} × 4.50 L × 2 = 31.957 mol = **32 mol**

ii) Calculating the number of moles of H^+ consumed when the discharge of the battery forms 138.1 g of $PbSO_4$.
n ($PbSO_4$) = (m ÷ M) = (138.1 g ÷ 303.26 g mol^{-1}) = 0.455 mol
n (H^+) = 2 × n ($PbSO_4$) = 2 × 0.455 = **0.911 mol**

iii) Calculating the concentration of H^+ in the discharged battery.
n (H^+) left over = 31.957 mol – 0.911 mol = 31.046 mol
c [H^+] = (n ÷ v) = (31.046 ÷ 4.5 L) = 6.899 = **6.9 mol L^{-1}**

iv) To show that the change in pH is negligible when the battery discharges.
The initial concentration of the H^+ ions = 2 × 3.55 mol L^{-1} = 7.1 mol L^{-1} (See section d, i)
The initial pH = –log [7.1] = –0.851
The final pH = –log [6.899] = –0.839. There is a very **small difference of 0.012 pH units.**

e) Hydrogen gas is flammable and oxygen supports combustion. The mixture can explode and cause fire hazards.

13.(2014:39) This is a straightforward empirical-molecular formula calculation.
a) The organic compound contains C, H, O and Br. The mass of the sample is 1.50 g.
m (CO_2) produced on combustion = 1.58 g
n (CO_2) produced = (1.58 ÷ 44.01) = 0.03590 mol
Therefore, n (C) = n (CO_2) = 0.03590 mol
Therefore, m (C) = 0.03590 × 12.01 = 0.4312 g. Therefore, carbon is **28.7%** of the compound.

m (H_2O) produced on combustion = 0.563 g.
n (H_2O) produced = (0.563 ÷ 18.016) = 0.03125 mol
Therefore n (H) produced = 2 × 0.03125 = 0.0625 mol
Therefore, m (H) produced = 0.0625 × 1.008 = 0.0630 g. Therefore, hydrogen is **4.2%** of the compound.

Another sample of mass 1.75 g results in 1.97 g of AgBr.
Therefore, the original mass, 1.50 g sample would have produced, [(1.50 × 1.97) ÷ 1.75] = 1.6886 g of AgBr.
Therefore, n (Br) = n (AgBr) = 1.6886 ÷ (107.9 + 79.90) = 0.008991 mol.
Therefore, m (Br) produced = 0.008991 × 79.90 = 0.7184 g. Therefore, bromine is **47.9%** of the compound.

Therefore, the mass of oxygen in the compound is = [1.50 – (0.4313 + 0.0630 + 0.7184)] = 0.2874 g. Oxygen is **19.2%** of the compound.
n (O) = (0.2874 ÷ 16.0) = 0.01796 mol.

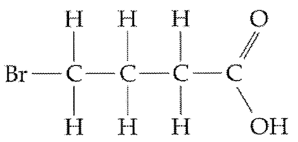
Calculation of the empirical formula

	28.7%	4.2%	19.2%	47.4%
Elements present	C	H	O	Br
Mass ratio (g)	0.04312	0.0630	0.2874	0.7184
Mole ratio	0.03590	0.0625	0.01796	0.008991
Simple ratio	3.993	6.951	1.998	1
Whole No. ratio	4	7	2	1

Therefore, the empirical formula is **$C_4H_7O_2Br$**

b) The empirical mass of the compound is = 48.04 + 7.056 + 32.0 + 79.9 = 167.0
Using the relationship, PV = nRT and [n = (PV ÷ RT)], the number of moles of the compound of mass 1.95 g is calculated as follows:
n = {(95.0 × 0.387) ÷ [8.314 × (273 + 105)]} = [36.765 ÷ (8.314 × 378)] = 0.01170 mol
Since 0.01170 mol of the compound has a mass of 1.95 g
1.00 mol of the compound (the molecular mass) is = (1.95 ÷ 0.01170) = 166.7 g mol^{-1}
As the empirical mass (167.0 g) and the molar mass (166.7 g) are equal within the limits of the experimental error, the molecular formula is the same as the empirical formula.
The molecular formula of the compound is = $\mathbf{C_4H_7O_2Br}$.

c) A possible structure for the formula is given below.



4-bromobutanoic acid

This compound could be 4-bromobutanoic acid. Other position isomers involving the –Br functional groups are also possible.

14.(2014:41) Rock phosphate is a non-renewable source of fertiliser. Because of the concern that it will run out eventually, urine is investigated as an alternative source of phosphate. This calculation deals with the analysis of urine as an alternative source.
a) Balanced equation for the hydrolysis of urea ($CO(NH_2)_2$) producing NH_3 and CO_2. (Hydrolysis is the decomposition of a compound with water.)

$CO(NH_2)_2(aq) + H_2O(\ell) \rightarrow 2\ NH_3(aq) + CO_2(g)$

b) Explanation: Why does the hydrolysis of urea cause an increase in pH?
Ammonia is produced during the hydrolysis of urea. Ammonia is further hydrolysed to produce OH^- ions. The reaction is:

$NH_3(aq) + H_2O(\ell) \rightarrow NH_4^+(aq) + OH^-(aq)$

The increase in the concentration of OH^- ions increases the pH of the solution.

c) Determining the concentration, in grams per litre of phosphorus in the urine.
Supplied: Molar masses: struvite = 245.418 g mol^{-1}; calcium hydroxyapatite = 1004.636 g mol^{-1}.
Note: The second paragraph of the question tells us that the phosphate in urine will precipitate as calcium hydroxyapatite ($Ca_{10}(PO_4)_6(OH)_2$), and struvite ($MgNH_4PO_4 \cdot 6\ H_2O$).

m ($MgNH_4PO_4 \cdot 6\ H_2O$) = 82.3% of the precipitate of 25.3 g = [(25.3 × 82.3) ÷ 100] = 20.823 g.
m ($Ca_{10}(PO_4)_8(OH)_2$) = 17.7% of the precipitate of 25.3 g = (25.3 – 20.823) = 4.478 g.
n (P) in struvite = n (struvite) = (20.823 ÷ 245.418) = 8.484 × 10^{-2} mol

n ($Ca_{10}(PO_4)_6(OH)_2$) = [(4.478 ÷ 1004.636) × 6] = 4.457 × 10^{-3} mol.
n (P) in $Ca_{10}(PO_4)_8(OH)_2$ × 6 = 4.457 × 10^{-2} × 6 = 2.674 × 10^{-2} mol

Total amount of P in both = 8.484 × 10^{-3} mol + 2.674 × 10^{-2} mol = 1.116 × 10^{-1} mol
Total mass of P in both = n × M = 1.116 × 10^{-1} mol × 30.97 = 3.456 g

The statement for part b) states the volume of urine taken for analysis is 5.00 L.
Therefore, the concentration of P in urine in g L^{-1} = (m ÷ v) = (3.456 ÷ 5.00) = 0.691 g L^{-1}.
= **6.91 × 10^{-1} g L^{-1}**

d) Since the concentration of the OH– ion increases, there is also an increase in the rate of collisions between the OH^- ions and the NH_4^+ ions. However, the rate of the forward reaction is higher than the rate of the reverse reaction which shifts the equilibrium towards the products which increases the yield of ammonia. See the following equation:

$NH_4^+(aq) + OH^-(aq) \rightleftharpoons NH_3(aq) + H_2O(\ell)$

e) The conversion of a liquid to a gas involves absorption of heat energy. Also, solubility of gases decreases with an increase in temperature. This releases ammonia as a gas. An increase in temperature favours gas formation and an equilibrium shift to the right.

f) Equation for the reaction of ammonia absorption by the acid to form ammonium sulfate.

$2\ NH_3(aq) + H_2SO_4(aq) \rightarrow 2\ NH_4^+(aq) + SO_4^{2-}(aq)$, or
$NH_3^+(aq) + HSO_4^-(aq) \rightarrow NH_4^+(aq) + SO_4^{2-}(aq)$